

Class 14: RNASeq Mini Project

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Background

Here, we will do RNA-seq analysis of lung fibroblasts with knockdown of the HOXA1 developmental transcription factor. We'll use DESeq for differential expression analysis, followed by pathway analysis using KEGG, GO, and Reactome.

Data Import

Read counts and metadata CSV files

```
# Load data files
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"
```

```
# Import and view metadata
colData = read.csv(metaFile, row.names=1)
colData
```

```
      condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
# Import and view countdata
countData = read.csv(countFile, row.names=1)
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
		SRR493371				
ENSG00000186092		0				
ENSG00000279928		0				
ENSG00000279457		46				
ENSG00000278566		0				
ENSG00000273547		0				
ENSG00000187634		258				

Q. Complete the code below to remove the troublesome first column from countData

```
# Remove the first column of countData
countData <- as.matrix(countData[,-1])
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46

ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q. Complete the code below to filter `countData` to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns). Tip: What will `rowSums()` of `countData` return and how could you use it in this context?

Since each row of `countData` corresponds to a gene, `rowSums()` will return a vector with the total number of reads per gene. Since we want to keep genes that do *not* have zero read counts, we can filter the rows of `countData` with `rowSums(countData) != 0` (returns TRUE for row indices that don't have a zero sum), as shown below.

```
# Make logical vector identifying rows that do NOT sum to 0
to.keep <- rowSums(countData) != 0

# Filter countData using the to.keep logical vector
countData = countData[to.keep,]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

```
# Check the new dimensions of countData
dim(countData)
```

```
[1] 15975      6
```

Sanity check

Let's make sure the column names of our `countdata` align with the row names of our metadata (i.e. the names of samples).

```
# Column names of countdata
colnames(countData)
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
# Row names of metadata  
rownames(colData)
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
# Check that the countdata column names are the same as the metadata row names.  
colnames(countData) == rownames(colData)
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

Setup DESeq object

```
library(DESeq2)
```

```
dds = DESeqDataSetFromMatrix(countData=countData, colData=colData, design=~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds
```

```
class: DESeqDataSet  
dim: 15975 6  
metadata(1): version  
assays(1): counts  
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345  
                  ENSG00000271254  
rowData names(0):  
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371  
colData names(1): condition
```

Run DESeq analysis pipeline

```
dds = DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

Extract the results

Big table with log2 fold changes and p-values, comparing HoxA1 knockdown (hoxa1_kd) versus control siRNA (control_sirna)

```
res = results(dds)
```

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

4349 genes are upregulated and 4396 genes are downregulated at the default 0.1 adjusted p-value cutoff.

```
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

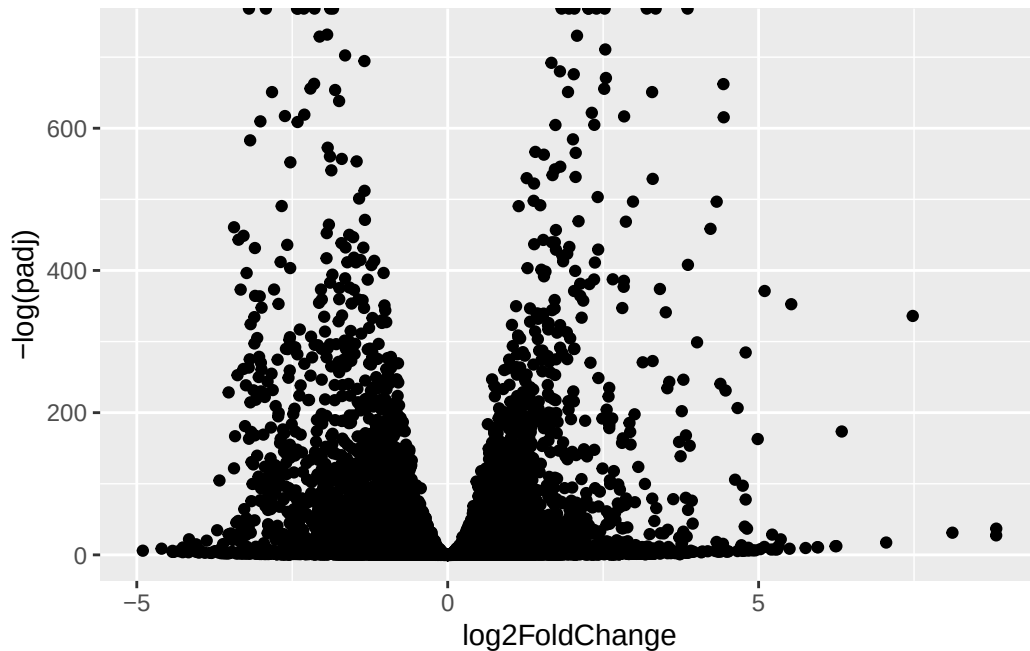
Data Viz

Make a volcano plot of $\log_2(\text{fold change})$ versus $-\log(\text{adjusted p-value})$

```
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).



Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

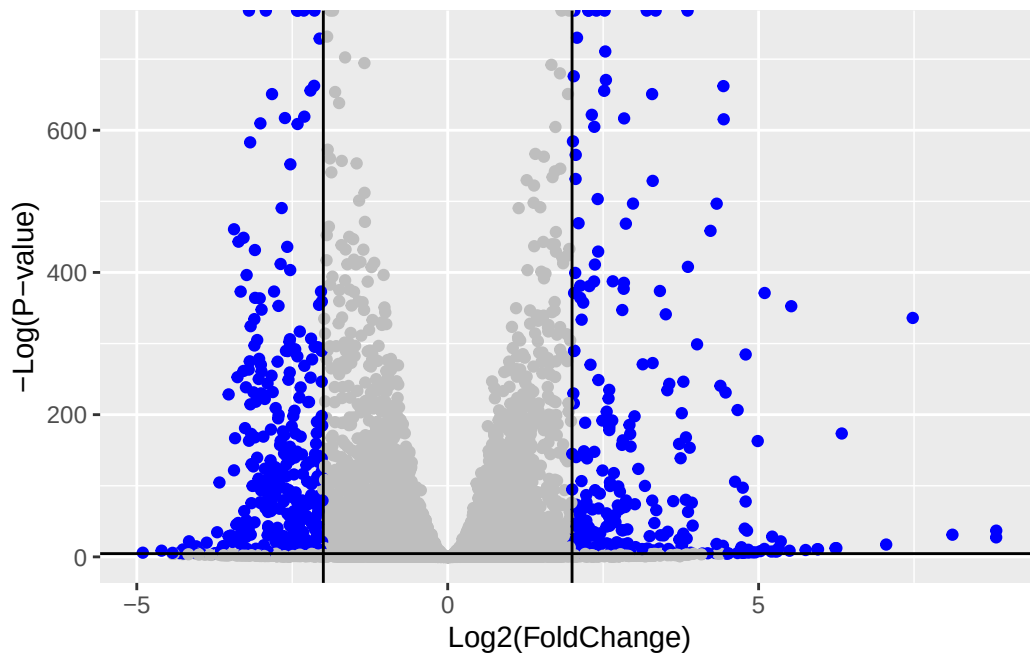
```
# Make a color vector for all genes
mycols <- rep("gray", nrow(res))

# Color blue the genes with |L2FC| > 2
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

# Color gray genes with adjusted p-value > 0.01
mycols[ res$padj > 0.01 ] <- "gray"

# Add custom color vector, axis labels, cutoff lines to plot
ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point(col = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log(0.01))
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Q. Volcano plot highlighting $\text{padj} < 0.05$ and $\text{abs}(\text{foldchanges}) > 2$.

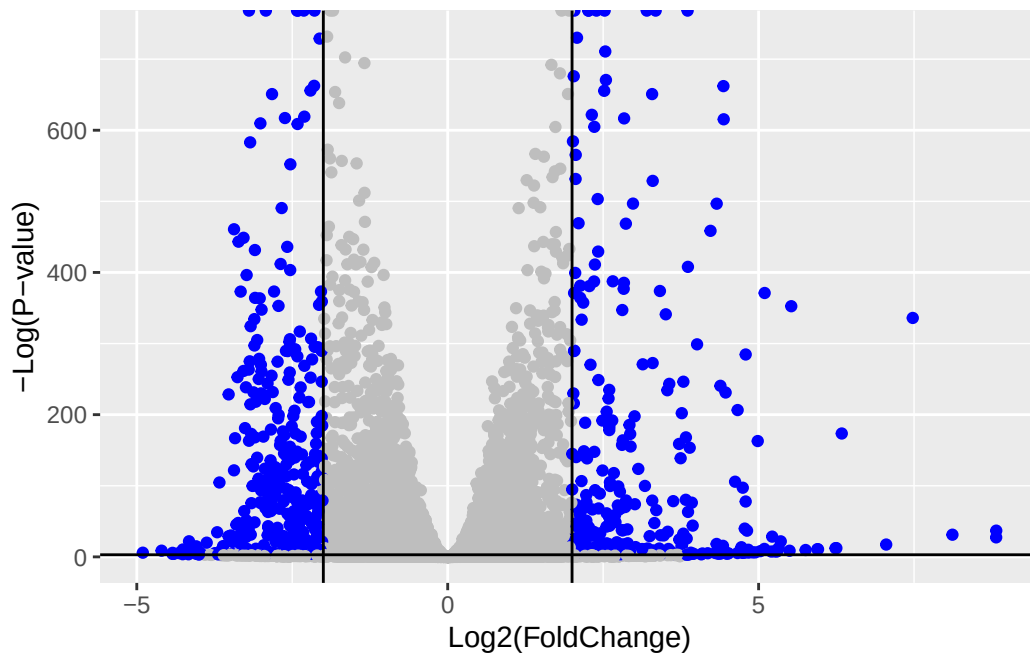
The following volcano plot has a less stringent adjusted p-value cutoff of 0.05 (the previous plot uses a cutoff of 0.01).

```
# Make a color vector for all genes
mycols_05 <- rep("gray", nrow(res))

# Color blue the genes with |L2FC| > 2 & padj < 0.05
mycols_05 [ abs(res$log2FoldChange) > 2 & res$padj < 0.05 ] <- "blue"

# Add custom color vector, axis labels, cutoff lines to plot
ggplot(res) +
  aes(log2FoldChange, -log(padj)) +
  geom_point(col = mycols_05) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log(0.05))
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Add gene annotation data

Add gene SYMBOL, ENTREZID, and GENENAME to `res` object.

Q. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
# Load AnnotationDbi package
library("AnnotationDbi")

# Load annotation data package for Homo sapiens
library("org.Hs.eg.db")
```

```
# Examine formats available in org.Hs.eg.db
columns(org.Hs.eg.db)
```

```
[1] "ACCNUM"      "ALIAS"       "ENSEMBL"     "ENSEMBLPROT" "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"      "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"    "GO"          "GOALL"        "IPI"          "MAP"
```

```
[16] "OMIM"          "ONTOLOGY"      "ONTOLOGYALL"   "PATH"          "PFAM"
[21] "PMID"          "PROSITE"       "REFSEQ"        "SYMBOL"        "UCSCKG"
[26] "UNIPROT"
```

```
# Add a new column to res with SYMBOL format of gene names
res$symbol = mapIds(org.Hs.eg.db,
  keys=row.names(res),
  keytype="ENSEMBL",
  column="SYMBOL",
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
# Add a new column to res with ENTREZID format of gene names
res$entrez = mapIds(org.Hs.eg.db,
  keys=row.names(res),
  keytype="ENSEMBL",
  column="ENTREZID",
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
# Add a new column to res with GENENAME format of gene names
res$name = mapIds(org.Hs.eg.db,
  keys=row.names(res),
  keytype="ENSEMBL",
  column="GENENAME",
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
# View annotated res object
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna
 Wald test p-value: condition hoxa1 kd vs control sirna
 DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248215	0.551863	5.81042e-01

ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51281e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02
ENSG00000188157	9128.439422	0.3899088	0.0467164	8.346302	7.04333e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez		name
	<numeric>	<character>	<character>		<character>
ENSG00000279457	6.86555e-01	NA	NA		NA
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.76553e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.21970e-16	AGRN	375790		agrin
ENSG00000237330	NA	RNF223	401934	ring finger protein ..	

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
# Reorder res by p-value
res = res[order(res$pvalue),]

# Save results to CSV file
write.csv(res, file="deseq_results.csv")
```

Pathway analysis

KEGG, GO, and REACTOME

KEGG

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)
```

```
# Load KEGG pathway data
```

```
data(kegg.sets.hs)
```

```
# Load vector of kegg.sets.hs indices corresponding to signaling/metabolic pathways
```

```
data(sigmet.idx.hs)
```

```
# Filter kegg.sets.hs to signaling and metabolic pathways only
```

```
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]
```

```
# Examine the first 3 pathways
```

```
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

\$`hsa00230 Purine metabolism`

[1]	"100"	"10201"	"10606"	"10621"	"10622"	"10623"	"107"	"10714"
[9]	"108"	"10846"	"109"	"111"	"11128"	"11164"	"112"	"113"
[17]	"114"	"115"	"122481"	"122622"	"124583"	"132"	"158"	"159"
[25]	"1633"	"171568"	"1716"	"196883"	"203"	"204"	"205"	"221823"
[33]	"2272"	"22978"	"23649"	"246721"	"25885"	"2618"	"26289"	"270"
[41]	"271"	"27115"	"272"	"2766"	"2977"	"2982"	"2983"	"2984"
[49]	"2986"	"2987"	"29922"	"3000"	"30833"	"30834"	"318"	"3251"
[57]	"353"	"3614"	"3615"	"3704"	"377841"	"471"	"4830"	"4831"
[65]	"4832"	"4833"	"4860"	"4881"	"4882"	"4907"	"50484"	"50940"
[73]	"51082"	"51251"	"51292"	"5136"	"5137"	"5138"	"5139"	"5140"
[81]	"5141"	"5142"	"5143"	"5144"	"5145"	"5146"	"5147"	"5148"
[89]	"5149"	"5150"	"5151"	"5152"	"5153"	"5158"	"5167"	"5169"
[97]	"51728"	"5198"	"5236"	"5313"	"5315"	"53343"	"54107"	"5422"
[105]	"5424"	"5425"	"5426"	"5427"	"5430"	"5431"	"5432"	"5433"
[113]	"5434"	"5435"	"5436"	"5437"	"5438"	"5439"	"5440"	"5441"
[121]	"5471"	"548644"	"55276"	"5557"	"5558"	"55703"	"55811"	"55821"
[129]	"5631"	"5634"	"56655"	"56953"	"56985"	"57804"	"58497"	"6240"
[137]	"6241"	"64425"	"646625"	"654364"	"661"	"7498"	"8382"	"84172"
[145]	"84265"	"84284"	"84618"	"8622"	"8654"	"87178"	"8833"	"9060"
[153]	"9061"	"93034"	"953"	"9533"	"954"	"955"	"956"	"957"
[161]	"9583"	"9615"						

Set up `foldchanges` vector with `log2foldchange` values from `res`, named by Entrez gene IDs from `res`:

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

1266	54855	1465	2034	2150	6659
-2.422719	3.201955	-2.313738	-1.888019	3.344508	2.392288

Run **gage** pathway analysis:

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)

# View results
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
# Look at the first few down (less) pathways
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

```
# Look at the first few up (greater) pathways
head(keggres$greater)
```

	p.geomean	stat.mean	p.val
hsa04640 Hematopoietic cell lineage	0.002822776	2.833362	0.002822776
hsa04630 Jak-STAT signaling pathway	0.005202070	2.585673	0.005202070
hsa00140 Steroid hormone biosynthesis	0.007255099	2.526744	0.007255099
hsa04142 Lysosome	0.010107392	2.338364	0.010107392
hsa04330 Notch signaling pathway	0.018747253	2.111725	0.018747253
hsa04916 Melanogenesis	0.019399766	2.081927	0.019399766

	q.val	set.size	exp1
hsa04640 Hematopoietic cell lineage	0.3893570	55	0.002822776
hsa04630 Jak-STAT signaling pathway	0.3893570	109	0.005202070
hsa00140 Steroid hormone biosynthesis	0.3893570	31	0.007255099
hsa04142 Lysosome	0.4068225	118	0.010107392
hsa04330 Notch signaling pathway	0.4391731	46	0.018747253
hsa04916 Melanogenesis	0.4391731	90	0.019399766

The top less/down pathway from `keggres` is “Cell cycle”, with the KEGG pathway id `hsa04110`. Make a plot of this pathway, with our RNA-Seq expression results overlaid:

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

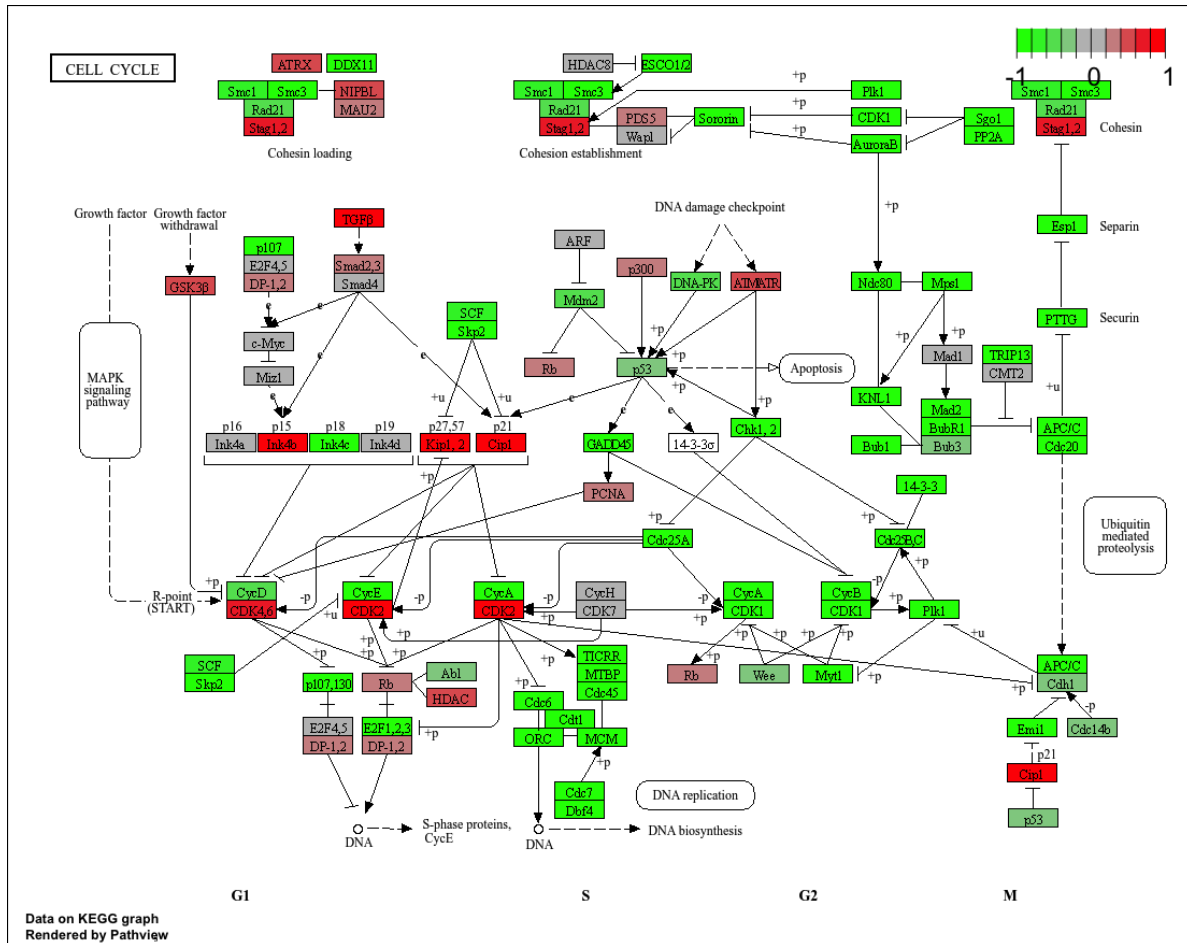


Figure 1: KEGG Cell cycle pathway, with RNA-Seq expression results in color

```
# We can also generate a PDF output of the same data
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

```
[,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"
```

Plot the top 5 upregulated KEGG pathways

```
## Access rownames for top 5 up/greater pathways
keggrespathways_up <- rownames(keggres$greater)[1:5]
# Extract the 8 character long IDs part of each string
keggresids_up = substr(keggrespathways_up, start=1, stop=8)
# View kegg ids
keggresids_up
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

Pass the vector of top 5 upregulated KEGG pathway ids to **pathview()**:

```
pathview(gene.data=foldchanges, pathway.id=keggresids_up, species="hsa")
```

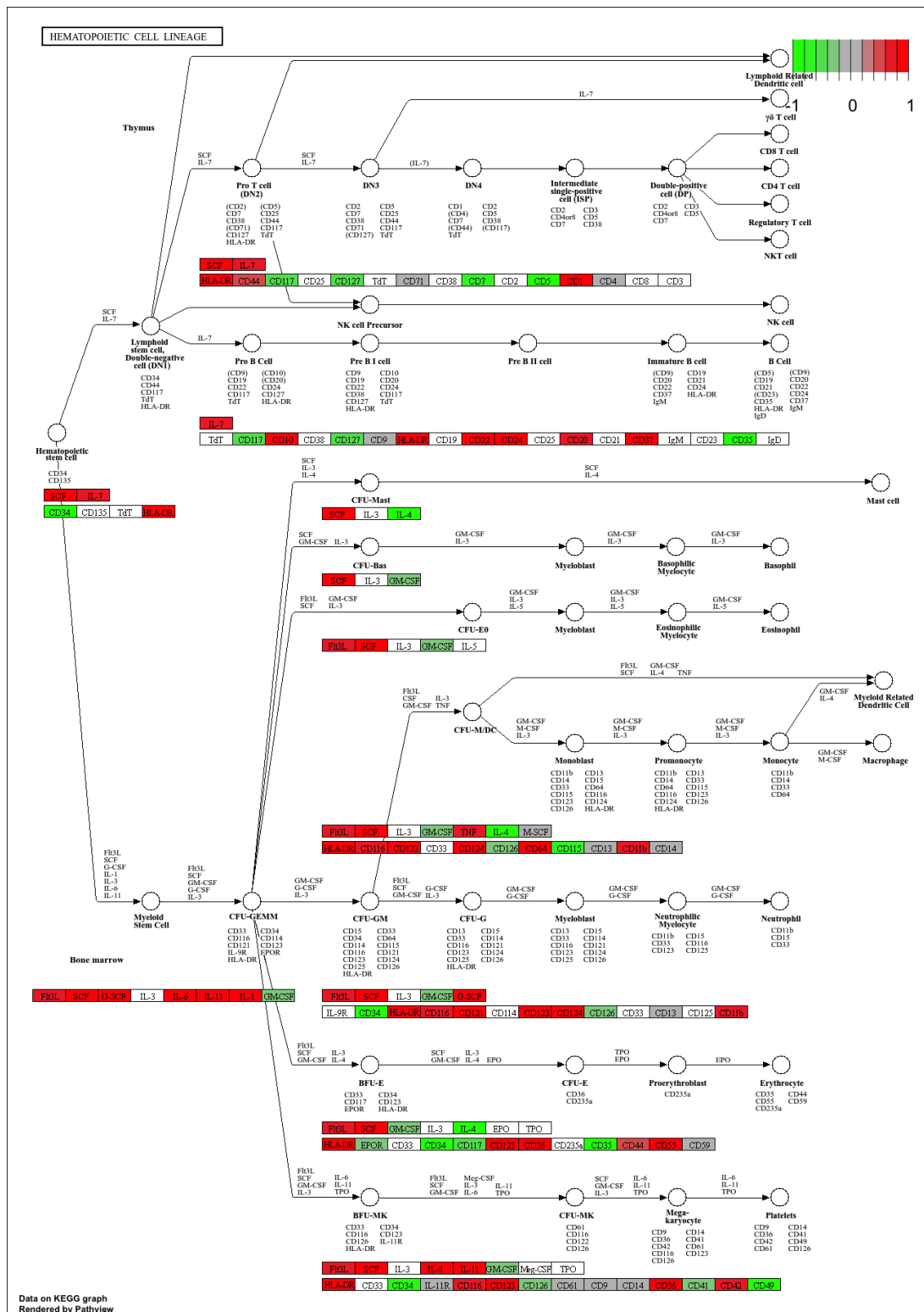



Figure 2: KEGG Hematopoietic cell lineage, with RNA-Seq expression results in color

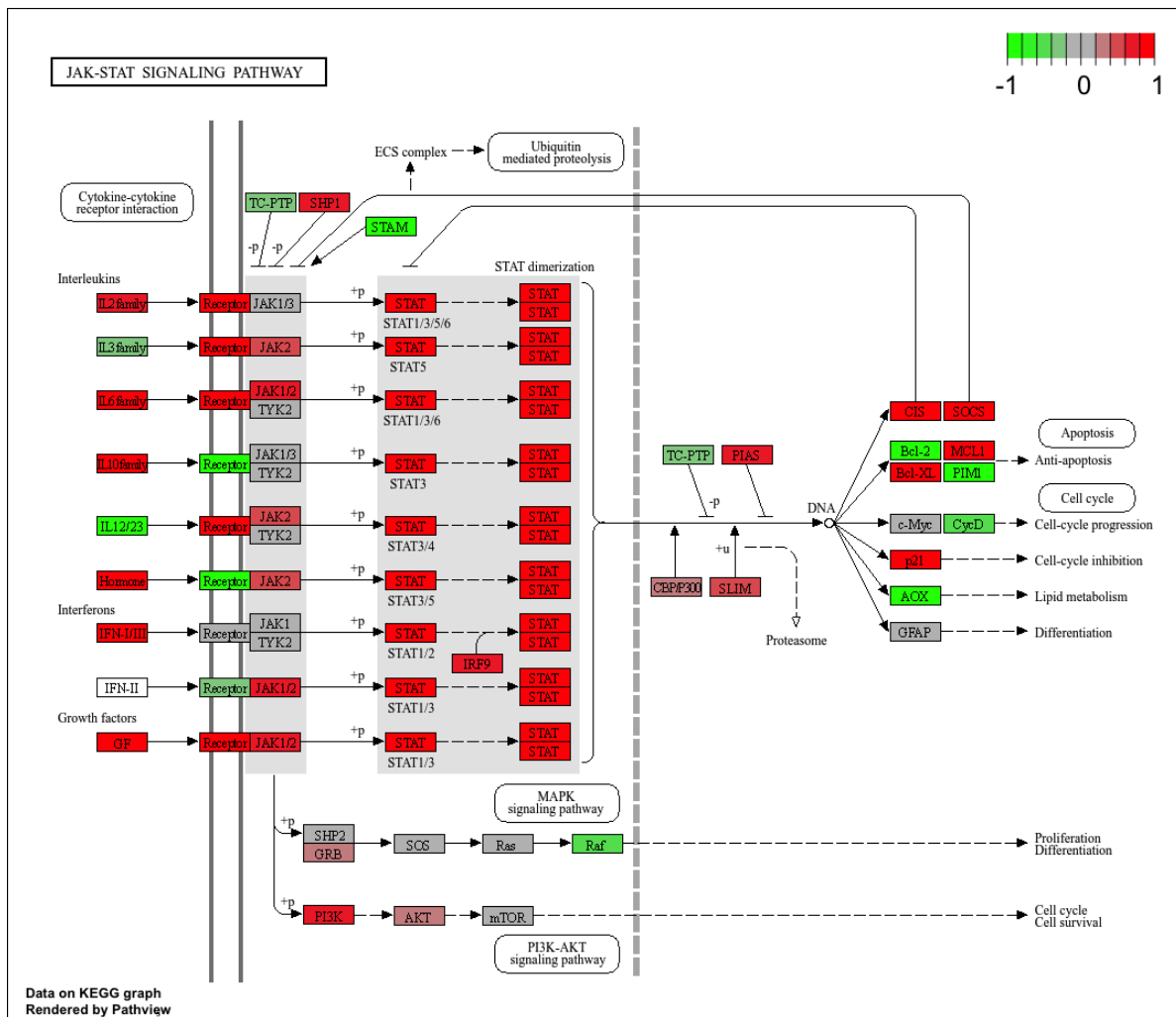


Figure 3: KEGG Jak-STAT signaling pathway, with RNA-Seq expression results in color

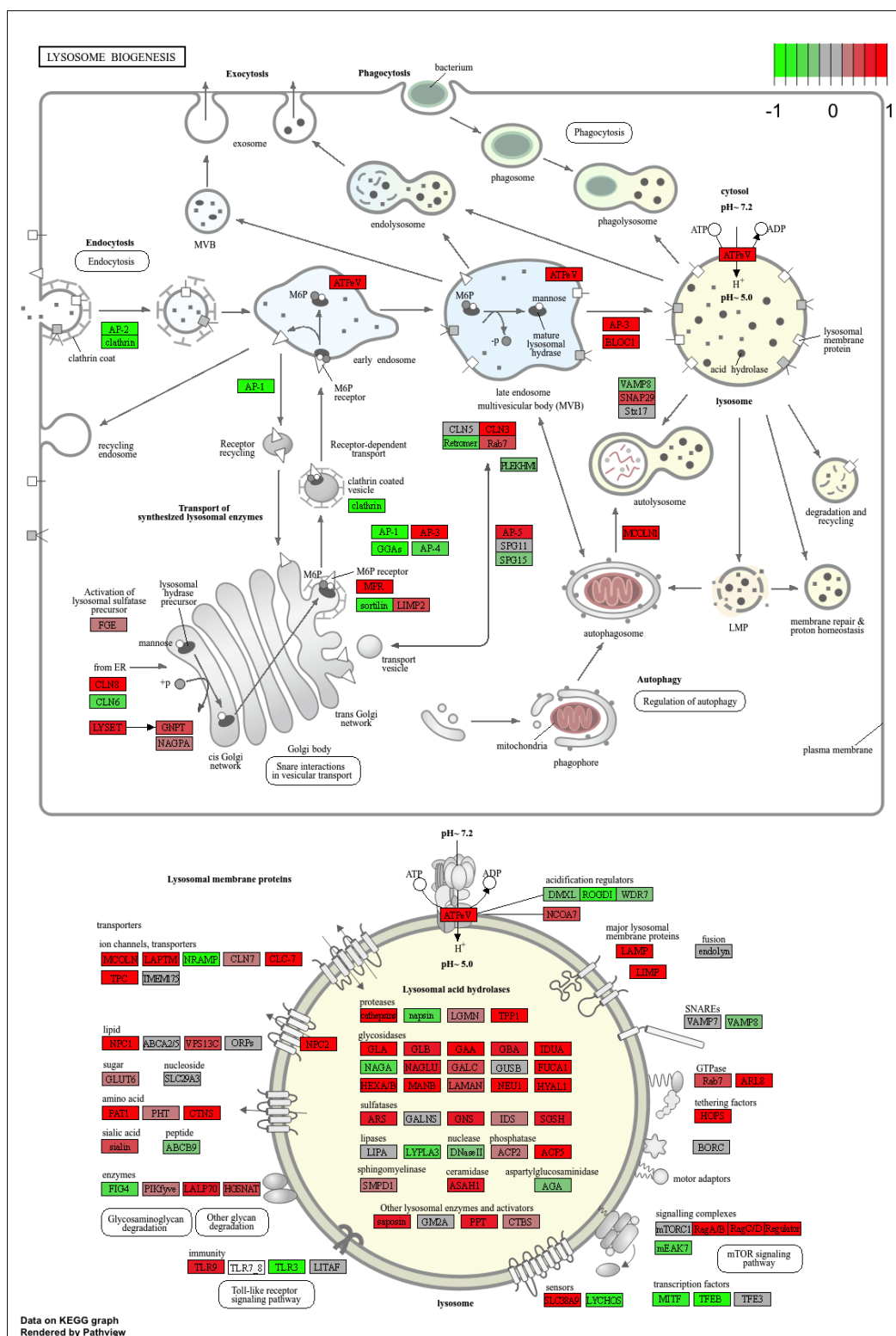


Figure 5: KEGG Lysosome, with RNA-Seq expression results in color

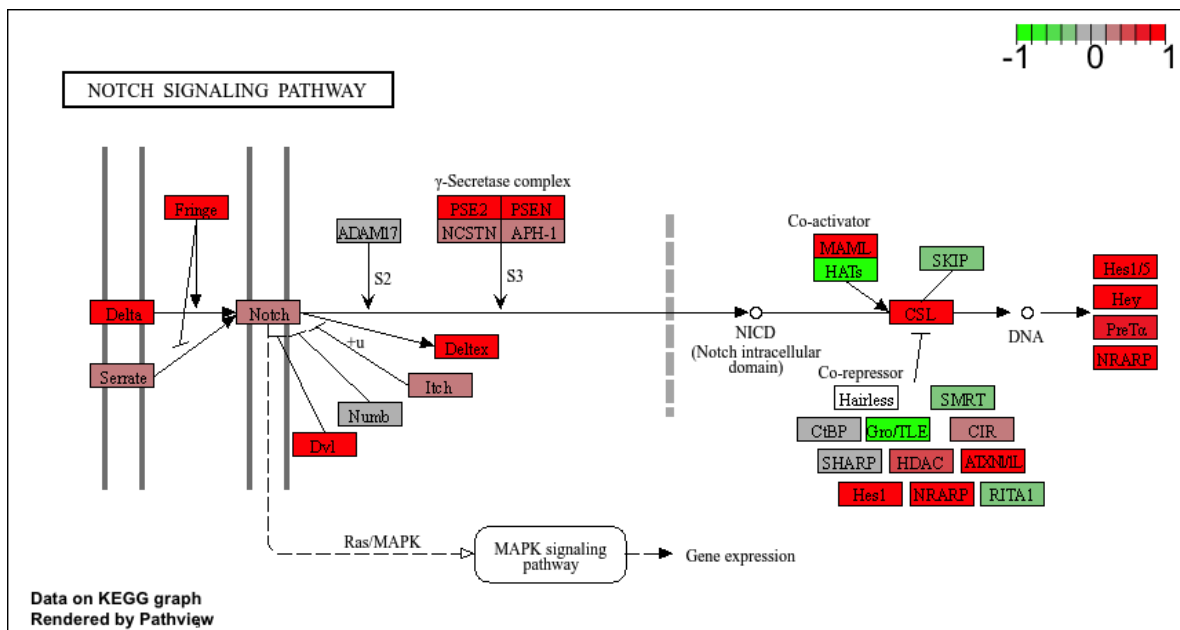


Figure 6: KEGG Notch signaling pathway, with RNA-Seq expression results in color

Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
## Access rownames for top 5 down/less pathways
keggrespathways_down <- rownames(keggres$less)[1:5]
# Extract the 8 character long IDs part of each string
keggresids_down = substr(keggrespathways_down, start=1, stop=8)
# View kegg ids
keggresids_down
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

Pass the vector of top 5 downregulated KEGG pathway ids to **pathview()**:

```
pathview(gene.data=foldchanges, pathway.id=keggresids_down, species="hsa")
```

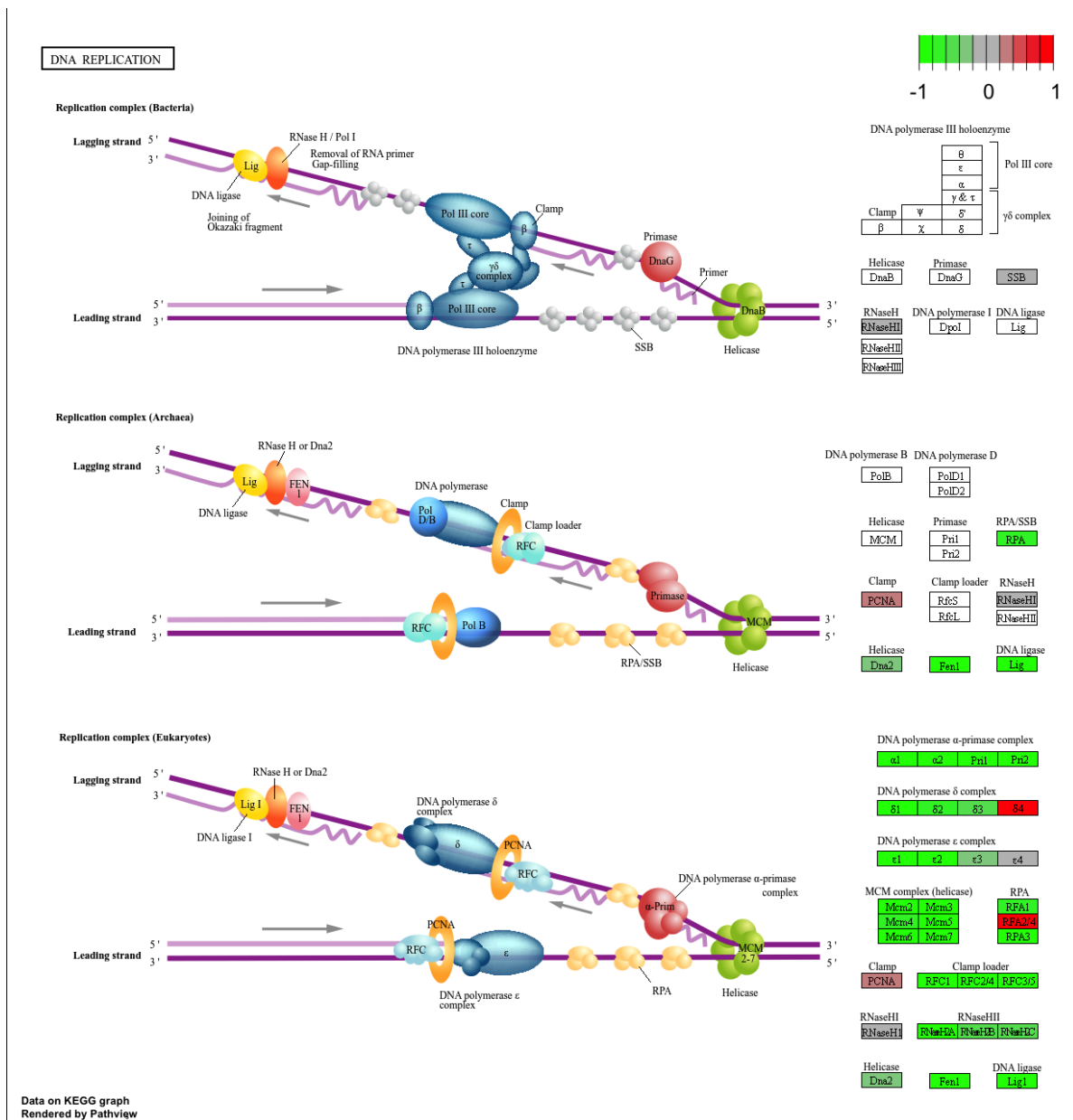



Figure 8: KEGG DNA replication, with RNA-Seq expression results in color

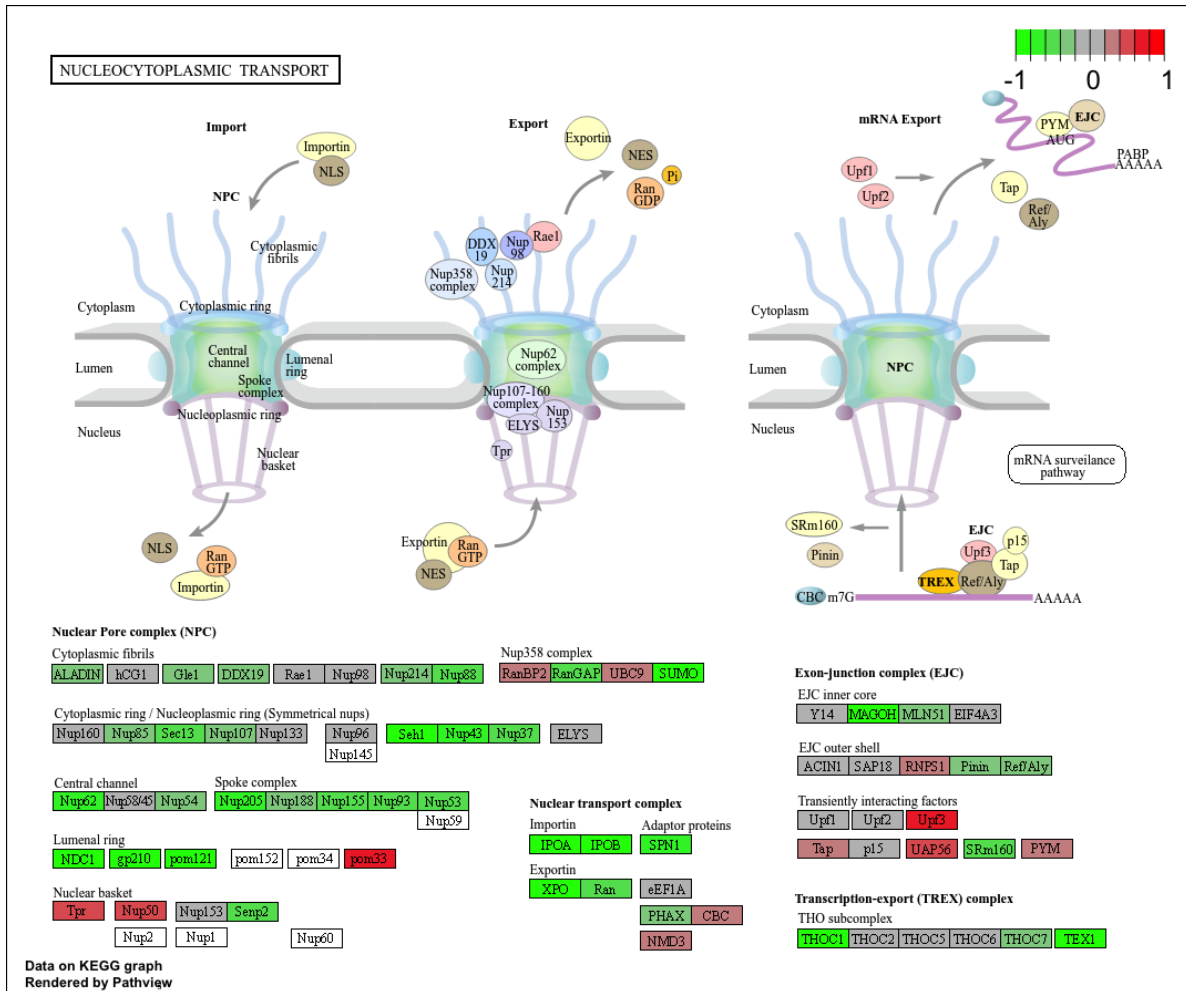


Figure 9: KEGG RNA transport, with RNA-Seq expression results in color

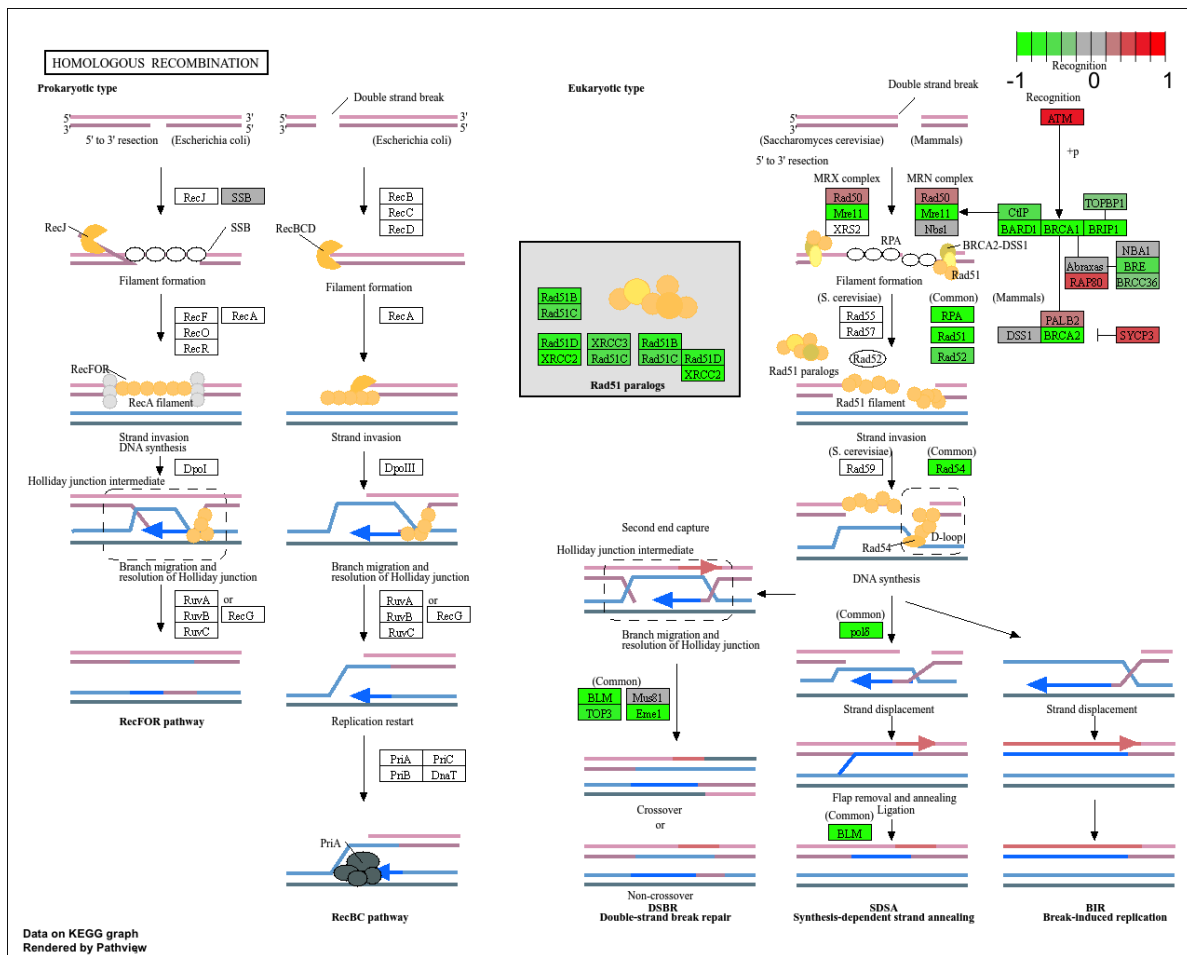


Figure 10: KEGG Homologous recombination, with RNA-Seq expression results in color

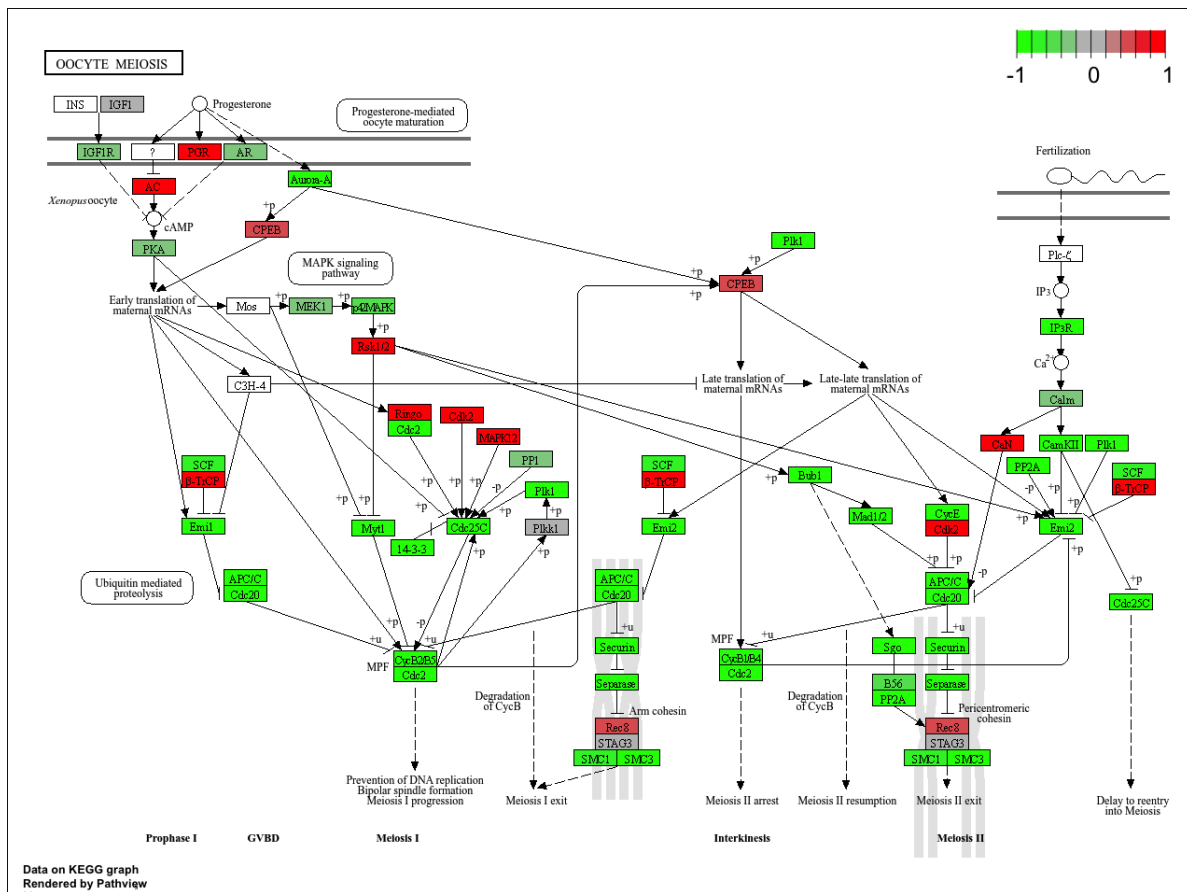


Figure 11: KEGG Oocyte meiosis, with RNA-Seq expression results in color

Gene Ontology

```
# Load GO terms
data(go.sets.hs)

# Load list of go.sets.hs indices corresponding to BP, CC, and MF ontologies
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

# Get the results
gobpres = gage(foldchanges, gsets=gobpsets)
```

```
# View the results
lapply(gobpres, head)
```

\$greater

	p.geomean	stat.mean	p.val
G0:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009 morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729 tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610 behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562 epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295 tube development	5.953254e-04	3.253665	5.953254e-04

	q.val	set.size	exp1
G0:0007156 homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009 morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729 tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610 behavior	0.1967577	426	1.925222e-04
G0:0060562 epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295 tube development	0.3565320	391	5.953254e-04

\$less

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10

	q.val	set.size	exp1
G0:0048285 organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280 nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067 mitosis	5.841698e-12	352	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059 chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236 mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

	stat.mean	exp1
G0:0007156 homophilic cell adhesion	3.824205	3.824205
G0:0002009 morphogenesis of an epithelium	3.653886	3.653886
G0:0048729 tissue morphogenesis	3.643242	3.643242
G0:0007610 behavior	3.565432	3.565432

G0:0060562	epithelial tube morphogenesis	3.261376	3.261376
G0:0035295	tube development	3.253665	3.253665

Reactome

There is an R package for this analysis and a new-ish website.

To use the website you can paste or upload a list of your DEGs.

```
# Extract significant genes from res
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]

# Print number of extracted genes
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
# Write list as a plain text file
write.table(sig_genes, file="significant_genes.txt",
            row.names=FALSE,
            col.names=FALSE,
            quote=FALSE)
```

Upload txt file to Reactome, selecting the parameter “Project to Humans”:

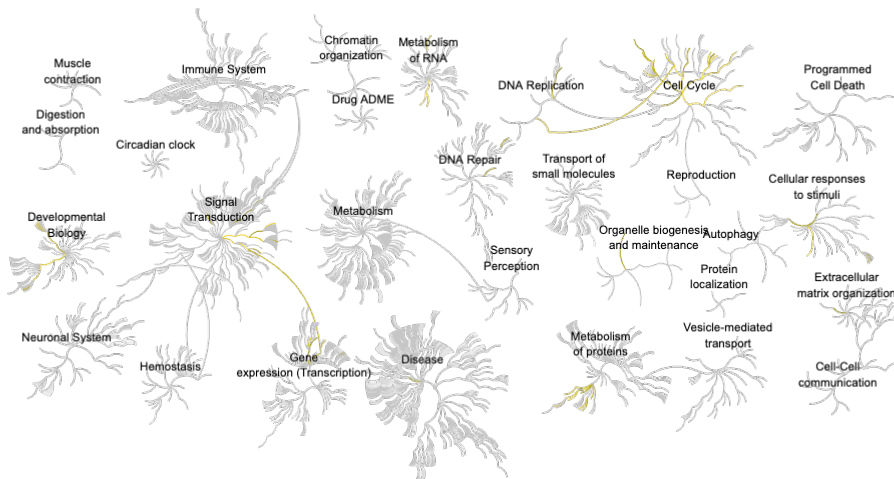


Figure 12: Reactome Pathways Overview

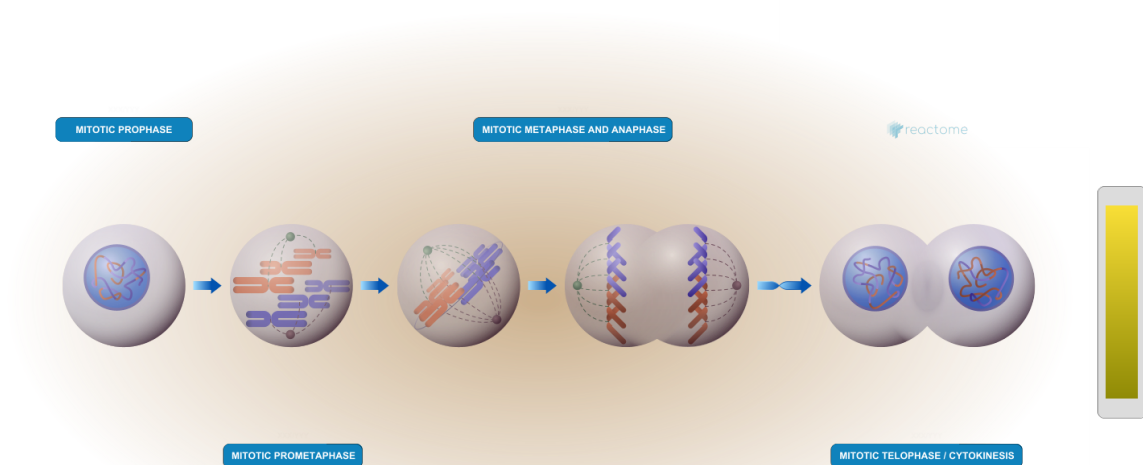


Figure 13: Reactome Cell Cycle M Phase Illustration

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway with the most significant “Entities p-value” is **Cell cycle** (with an Entities pValue of 2.63E-5).

Generally speaking, the most significant pathways outputted by Reactome are found in the “Cell cycle, Mitotic” and “Cell cycle, Checkpoints” branches. This matches the KEGG results, which named “Cell cycle” as the top downregulated (“less”) pathway.

One factor for differences in KEGG and Reactome results is how the two define and map pathways, which affects the composition, size, and specificity of individual pathways. A given KEGG pathway will not necessarily be a one-to-one match with its counterpart/most closely related pathway in Reactome. Additionally, KEGG and Reactome have different areas of emphasis: while KEGG pathways tend to be broad and systems-level, Reactome focuses more on mechanisms/signaling. Another factor is the difference in the number/types of annotated genes available for mapping in each method.

Save our results

See end of “Add gene annotation data” section.